Bogleheads

Importance of saving early

Saving early is important, as the power of *compound interest* will help you to reach your savings goal, even with low expected returns. The concept applies for savings towards any goal, such as retirement, purchasing a home, or saving for an education. The longer the timeframe, the larger the impact of the power of compounding.

If you start late, you can still catch up. However, you will need to invest more money, as compounding has much less time to have an effect.

> Compound interest is the eighth wonder of the world. He who understands it, earns it ... he who doesn't ... pays it.[1]
>
> — Albert Einstein

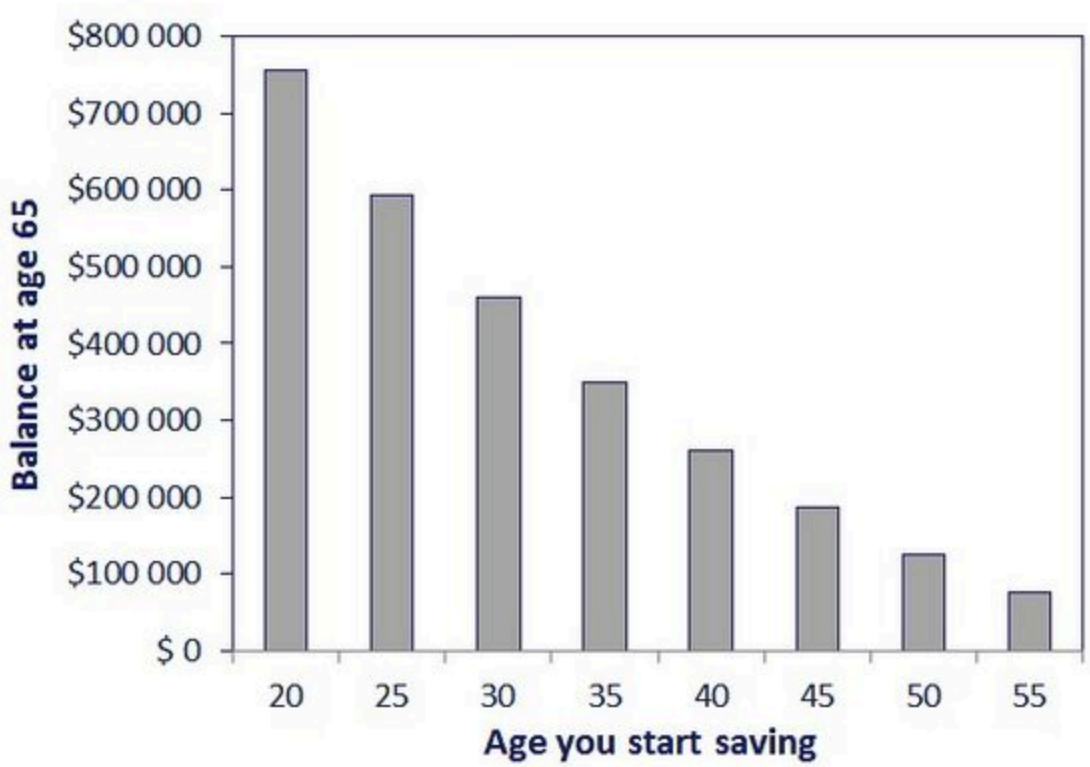
Compound interest

See also: Compound interest - Wikipedia

Compound interest is interest added to the principal of a deposit or loan so that the *added interest* also earns interest from then on. This addition of interest to the principal is called *compounding*. A bank account, for example, may have its interest compounded every year: in this case, an account with $1,000 initial principal and 5% interest per year would have a balance of $1,050 at the end of the first year, $1,102.50 at the end of the second year, $1,157.63 at the end of the third year, and so on.

How much more will you get if you start early?

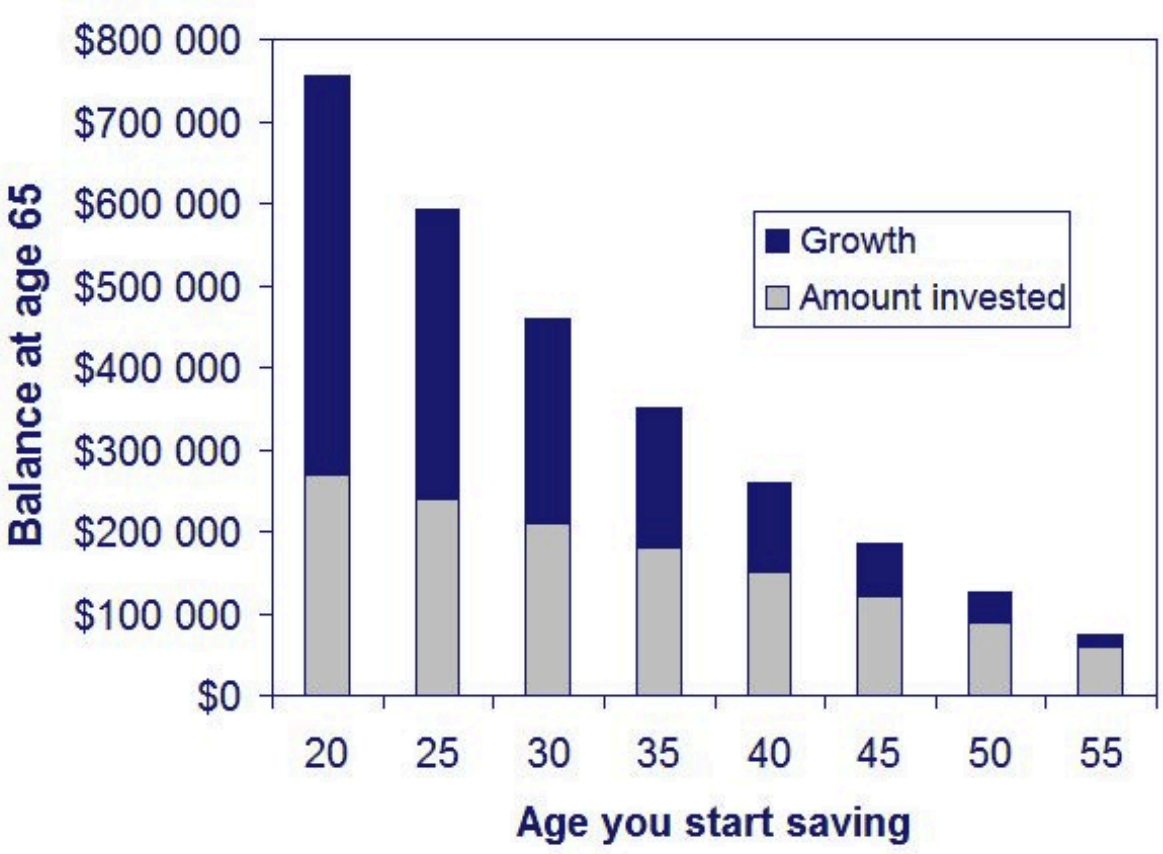
If you invest $6,000 at the beginning of each year[2] (the equivalent of $500 per month, but placed at the beginning of the year for easier calculations), and start at age 55, assuming 4% annualized nominal return,[note 1] your balance at age 65 will be about $75,000. If you start at age 40, you will end up with about $260,000. And if you start at age 25, your balance at age 65 will be about $593,000.[note 2] The graph below illustrates this.



To better understand what is going on, we can separate the amount invested in total from the growth of the invested money at 4% interest. Refer to the figure below.

Going from right to left, the lower (gray) bars represent your contributions. The sooner you start to invest, the more money you will have at retirement.

Now, look at the top (black) bars. Going from right to left, the sooner you start to invest, compounded interest has more time to work. Stated another way, the effect of compounding at age 20 will contribute much more significantly to your savings than contributions made later on.

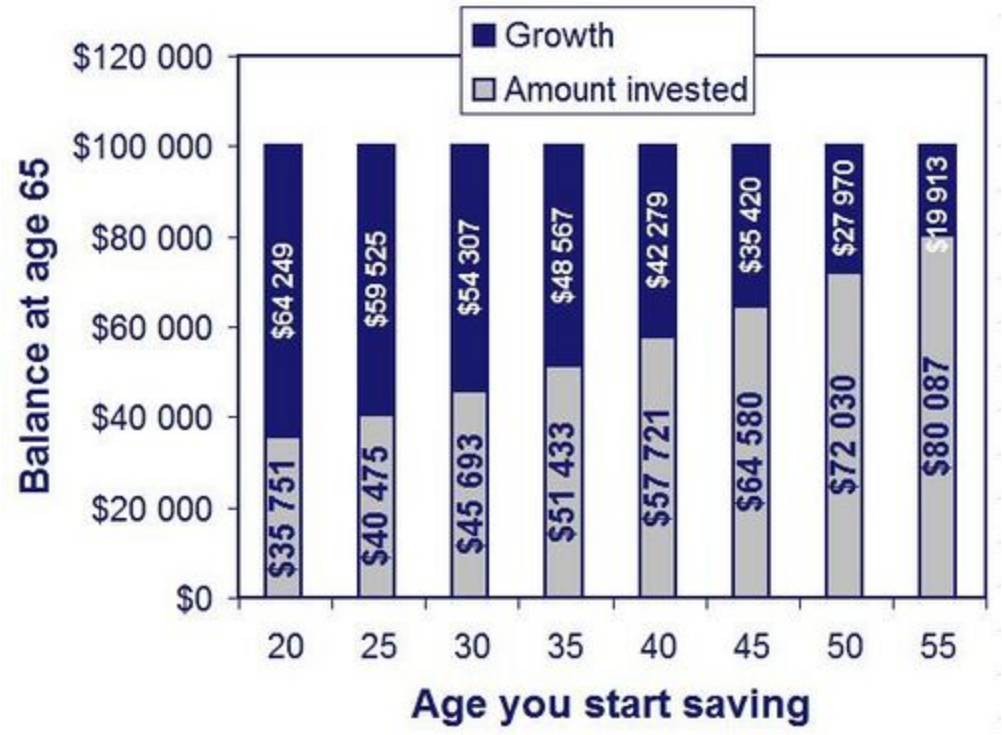


What does it take to catch up if you start late?

Suppose you want $100,000 at age 65, and the interest rate is again 4%. If you start at age 25, you will need to save only about $1,000 a year. At age 40, you will need to save about $2,300 a year. And if you start at age 55, the amount needed is over $8,000 per year.



The total amount needed to reach $100,000 at age 65 also increases as the saving period gets shorter:



See also

- Comparing investments - The "basics" on the time value of money.

- Invest early and often (Rule #2) - This short video illustrates the miracle of compound interest and the importance of starting to save early with a simple example of two young college graduates.

Notes

1. The 114-year historical record of global stock and bond returns suggests that a globally balanced stock/bond portfolio will likely provide an annualized compound real return between 3% and 4%, assuming minimal expenses and no tax expense. See Global investment portfolios 2015 update (https://blbarnitz4.wordpress.com/2015/02/12/global-investment-portfolios-2015-update/), from our site's unofficial (transferred) blog.
2. Calculations are done as:
   Starting at age 25: 592,959.22 = FV(4%, 65-25, -6000,0,1)
   The last parameter of FV has Type set to 1, which represents a payment due at the beginning of the period.

References

1. "Quote by Albert Einstein: "Compound interest is ..." " (https://www.goodreads.com/quotes/76863-compound-interest-is-the-eighth-wonder-of-the-world-he). Retrieved February 4, 2023.
2. Charu Chander Gross (December 12, 2011). "Need proof?" (https://web.archive.org/web/20141128141326/http://vanguardblog.com:80/2011/12/12/need-proof/). Vanguard Blog. Archived from the original (http://vanguardblog.com:80/2011/12/12/need-proof/) on 2014-11-28. Retrieved 2023-02-04.

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